

Ashutosh Verma [7+ Years]

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Education

Bachelor of Technology (Computer Science)

Gurukul Kangri University : June 2014 – May 2018

Skills

Core Languages & Frameworks

- **Python** – 7+ years (Advanced: OOP, async, decorators, multiprocessing, profiling, performance tuning)
 - **FastAPI** – 6 years (async APIs, background tasks, dependency injection, Pydantic validation)
 - **Django / Django REST Framework** – 6 years (MVT, ORM, DRF, Celery integration)
 - **Flask** – 3 years (lightweight services, internal tooling)
- **JavaScript / TypeScript** – 6 years
 - **React / Next.js** – 5 years (hooks, SSR/SSG/ISR, PWA, SEO, performance optimization)
 - **Node.js / Express** – 4 years (API gateways, orchestration, workers)
 - **TypeScript** – 4 years (strict typing in React/Node stacks)
- **Go (Golang)** – 3 years (Accounts API optimization, concurrency, REST services)
- **Java** – 3 years (Android Native apps, APIs)
- **Swift** – 2–3 years (iOS native apps)
- **C++** – 3 years (MediaPipe AR engine, WASM, performance-critical modules)
- **C# / .NET** – 1.5 years (legacy refactor, Windows services, SQL Server integration)

Systems & Architecture

- **System Design (LLD/HLD)** – 6 years
- **Microservices & Distributed Systems** – 6 years (HA clusters, replication, service orchestration)
- **API Protocols** – REST, GraphQL, gRPC – 6 years
- **Event-Driven Architectures** – 4+ years (Kafka, RabbitMQ, Pub/Sub, SQS)
- **Scalability Patterns** – Load balancing, caching layers, sharding, replication, saga patterns

Databases & Storage

- **Relational:** PostgreSQL (6), MySQL (6), SQL Server (3)
- **NoSQL:** MongoDB (6), DynamoDB (4), Cassandra (4)
- **Caching:** Redis (6), Memcached (2)
- **Search & Vector DBs:** Elasticsearch (4), OpenSearch (3), Pinecone (2, vector search POC)
- **Graph DB:** Neo4j (3)
- **Time-series DBs:** InfluxDB (3), TimescaleDB (3), Prometheus TSDB (4)
- **OLAP / Warehousing:** Snowflake (2), BigQuery (2), Redshift (2, integrations)
- **Replication & Recovery:** WAL, point-in-time recovery, HA setups

Messaging, Streaming & Workflow Orchestration

- **Kafka** – 5 years (streams, topics, consumer groups, backpressure handling)
- **RabbitMQ** – 5 years (pub-sub, task queues, retry strategies)
- **AWS SQS** – 5 years (queue-based workflows)
- **Google Pub/Sub** – 4 years
- **Workflow Engines:** Kestra, Airbyte, Airflow – 3 years (batch & streaming data pipelines)
- **Celery** – 5 years (task orchestration, retries, distributed workers)

Cloud & DevOps

- **Cloud Platforms:**
 - AWS – 5 years (EC2, S3, Lambda, RDS, DynamoDB, CloudWatch, API Gateway, ECS)
 - GCP – 5 years (GKE, Pub/Sub, BigQuery, Cloud Functions, Logging/Monitoring)
 - Azure – 4 years (App Services, SQL Server, VM hosting)
- **Containers & Orchestration:** Docker (6), Kubernetes (6), Helm (5), Docker Swarm (3)
- **Infra as Code:** Terraform (4), Ansible (4), Pulumi (2)
- **CI/CD Pipelines:** GitLab CI (6), Jenkins (5), GitHub Actions (3), ArgoCD (2)
- **Package Managers:** npm, Yarn, pnpm, Bun (3–5 each)

Monitoring, Observability & Security

- **Monitoring / Metrics:** Prometheus (5), cAdvisor (2), Node Exporter (2), CloudWatch (5)
- **Dashboards:** Grafana (5), Kibana (5)
- **Logging:** ELK Stack (5), Graylog (3), Loki (2)
- **Tracing:** Jaeger (3), OpenTelemetry (2), Sentry (4)
- **APM Tools:** NewRelic (4), Datadog (4)
- **Security:** OAuth2 (5), JWT (5), RBAC (5), API Gateway security, TLS/mTLS (3)

Frontend / Mobile Development

- **Web Frontend:** React (5), Next.js (5), Angular (2), Vanilla JS/HTML/CSS (6)
- **Mobile:**
 - React Native (2)
 - Native Android – Java (3)
 - Native iOS – Swift (2–3)
- **AR/VR:** MediaPipe (3), WebGL (2), WASM (2), ARCore/ARKit basics

AI, ML & Search

- **LLMs & Chatbots:** OpenAI APIs, Ollama, LangChain – 2 years
- **Search & NLP:** Semantic search, embeddings, vector DB integration – 2 years
- **Recommendation Engines:** Elastic-based and rule-based recommenders – 2 years
- **Profiling / Analytics:** Real-time user journey monitoring, sales funnel tracking – 3 years

Other Tools & Utilities

- **Version Control:** Git, GitLab, GitHub (7+)
- **Project Management:** Jira, Confluence, Trello (5+)
- **Testing Frameworks:** Pytest, Tox, Jest, Mocha, Chai, ts-jest (5)
- **Code Quality:** Husky, ESLint, Prettier (3+)

Leadership & Collaboration

- **Team Leadership & Mentorship:** 4+ years (led 5 engineers, onboarding juniors, reviewing code)
- **Cross-functional Collaboration:** Worked with Product, Business, and Design to resolve trade-offs & deliver impact.
- **Process Improvement:** Introduced TDD, CI/CD pipelines, observability culture, and coding standards across teams.

My Projects

NAV India (5 Aug 2024 – Present)

Accounts API Optimization

- **Tech Stack:** Golang, REST APIs, DB Optimization, Profiling, Caching

- **About:** Transformed Accounts API performance from **p90: 2s, p99: 60s** to **p95: 2ms, p99: 40–50ms**. Applied deep profiling, query optimization, and caching strategies to deliver extreme-scale performance.

Optimize Data Fetch from Exchanges

- **Tech Stack:** FastAPI, Python AsyncIO, Concurrency (10k+ jobs), Caching, Profiling
- **About:** Cut job runtimes from **200s** → **5–9s**; scaled system to reliably handle **~10k concurrent jobs** and **~500k API requests per minute** with robust async pipelines.

Python Version Upgrade

- **Tech Stack:** Python 3.x, Dependency Refactor, CI/CD, Test Automation
- **About:** Upgraded legacy Python services; resolved deprecated APIs, migrated dependencies, and hardened CI/CD pipelines with automated unit/integration testing.

Full-stack Observability & Health Monitoring

- **Tech Stack:** Grafana, Prometheus, Kibana, Jaeger, Graylog, Elasticsearch, cAdvisor, Node Exporter, Docker
- **About:** Implemented a **health monitoring agent** across all services and infra components. Integrated system + container exporters with central log aggregation, tracing, and dashboards. Reduced MTTR and improved reliability of distributed workloads.

.NET Legacy Code Refactor (incl. Data Sync Re-engineer)

- **Tech Stack:** .NET, SQL Server (Stored Procedures, ETL), Windows Services, Design Patterns (Repository, Factory, Dependency Injection)
- **About:** Re-architected legacy .NET data synchronization pipelines. Applied design patterns to improve throughput, maintainability, and reliability of critical workflows.

ShyftLabs (1 Nov 2023 – 5 Aug 2024)

FastAPI → TDD Migration

- **Tech Stack:** FastAPI, Python, SQLite (test harness), Pytest, Tox, GitLab CI/CD
- **About:** Converted FastAPI project to strict **TDD**; achieved **98%+ coverage** and uncovered **90+ critical bugs** before production.

CI/CD with Tox and GitLab Pipelines

- **Tech Stack:** Tox, GitLab CI/CD, Docker
- **About:** Automated build, test, and deployment pipelines for consistent, reliable delivery.

Pricing Change Implementation (Petco)

- **Tech Stack:** Python, SQL, API Development
- **About:** Delivered pricing-change engine for **offline US pet stores (Petco)**; integrated APIs with enterprise retail systems.

Data Pipelines for Event Logs

- **Tech Stack:** Python, MongoDB, Kestra, Airbyte, Airflow, Batch & Streaming Infra
- **About:** Built fault-tolerant pipelines to process **billions of events daily** using hybrid batch/streaming orchestration frameworks.

AI Chatbot & Semantic Search

- **Tech Stack:** FastAPI, Elasticsearch, Vector DB, Transformers, OpenAI/Ollama
 - **About:** Developed a **retrieval-augmented chatbot** using semantic embeddings and vector search, enabling intelligent contextual Q&A.
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ORNAZ (1 June 2018 – 1 Nov 2023)

Desktop E-commerce Replatform (PWA)

- **Tech Stack:** Next.js (SSG/SSR/ISR), Node.js, Nginx, CDN
- **About:** Built PWA with **Lighthouse 100/100** and **<0.5s load times**. Integrated multiple payment gateways and improved SEO/UX drastically.

Monolith → Microservices Migration

- **Tech Stack:** FastAPI, Django, Docker Swarm, gRPC/REST, PostgreSQL (HA 2M/3S), Celery, RabbitMQ
- **About:** Migrated monolith into **4 core services + workers** with HA database clusters. Achieved **99.99% uptime** with Docker Swarm orchestration.

Observability Platform

- **Tech Stack:** Grafana, Prometheus, ELK Stack, Jaeger, Sentry
- **About:** Built centralized monitoring for metrics, logs, and traces. Provided single-pane observability across microservices and APIs.

Diamond Compare Tool

- **Tech Stack:** Next.js, FastAPI, PostgreSQL, Elasticsearch, SSR/ISR
- **About:** Created diamond comparison app, increasing engagement **+35%** and conversion **+20%** with real-time indexed search.

Campaign Orchestrator

- **Tech Stack:** Node.js, Next.js, Redis, Webhooks, Email/SMS APIs
- **About:** Built trigger-based orchestration engine for campaigns; improved conversions with real-time customer journeys.

Virtual Try-On (Cross-Platform Native)

- **Tech Stack:** Java (Android), Swift (iOS), C++ (MediaPipe AR Engine), Web (JS/WebGL + WASM)
- **About:** Built AR try-on apps for **Android, iOS, and Web** with 90% shared C++ AR core. Delivered real-time jewelry trial; boosted engagement **50%** and sales conversion **30%**.

Virtual Trial Room (Web)

- **Tech Stack:** JavaScript/WebGL, MediaPipe (C++), WASM, CDN, SSR
- **About:** Launched browser-based AR trial room with static fallback for low-end devices. Reduced load times **80%**, increased engagement **30–40%**.

Mobile Apps (React Native)

- **Tech Stack:** React Native, REST/GraphQL APIs, Multi-Payment Integration
- **About:** Built iOS + Android apps using React Native. Cut dev cycle **30%**, increased reach **40%**.

CRM + Journey Analytics:

- **Tech Stack:** Django, PostgreSQL, Celery, Grafana, Prometheus, Real-time Session Monitoring
- **About:** Implemented CRM + per-session analytics system. Improved sales efficiency **40%** and reduced lead response times.

E-commerce Core (Ornaz v1) :

- **Tech Stack:** Django (MVT), PostgreSQL, Redis, RabbitMQ, Celery, HTML, Jinja2, JavaScript, PayU, Razorpay
- **About:** Mobile-first e-commerce platform. Increased mobile sales **40%**, reduced cart drop-offs **25%**, and boosted returning users.